1 CSE351 - Project #1: Movie Revenue Prediction

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Contributions

Hochan Jun

- Preprocessing (Cleaning the dataset) **numerical values**
- Solved 2 EDA problems: the movie genre trend shifting patterns, ROI & Analysis
- Modeling: simple linear regression model - testing and training + visualization
- Analysis on Case 1: Splitting data by the decades (70% training, 30% testing) => **before 2010 / after 2010 (including 2010)**
- Drawing a conclusion
- Presentation slides

Woochul Lim

- Preprocessing (Cleaning the dataset) **non-numerical values**
- Extracted the number of crews, the average revenue per country and company from non-neumeric data.
- Solved 2 EDA problems: the number of movies released by time periods, Strongest and weakest feautures correlated with Revenue.
- Modeling: multiple linear regression model, polynomial ridge regression - testing and training + visualization
- Analysis on Case 2: Splitting data by **random** (70% training, 30% testing)
- Summarizing RSE results
- Documentation

In [115…

```
# import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import ast
```

In [116…

```
# read the data
movies = pd.read_csv('./movie/tmdb_5000_movies.csv')
credits = pd.read_csv('./movie/tmdb_5000_credits.csv')
```

2 Preprocessing (Non-numeric Values)

Handling missing or invalid attributes in the data is essential for improving performace of models.
Several features in the original data are non-numerical and cannot be imputed using statistical methods such as mean, median, and random sampling, or imputation based on linear regression. Therefore these rows must be excluded.

The following attributes will be handled as dealing with non-numerical features:

In 'tmdb_5000_movies.csv'

- genres
- original_language
- prouction_companies
- production_countries
- release_date
- status

In tmdb_5000_credits.csv:

- cast
- crew

Since these features are not numeric, several imputation methods mentioned above are not applicable.
Therefore, rows containing invalid values will be excluded while training models.

Unreleased movies (status as 'unreleased' or 'post production') will also be excluded since these movies do not have revenue to train models.

In [117…

```
########## tmdb_5000_movies.csv ##########

# no movies with genres of NaN
movies[movies['genres'].isna()]

# Movies with empty genres will be excluded while traning models.
# This is not a numerical value so the imputation methods are not available.
movies = movies[movies['genres'] != '[]']

# no movies with original_language of NaN
movies[movies['original_language'].isna()]

# no movies with original_language of NaN
movies[movies['production_companies'].isna()]

# Movies with empty production_companies will be excluded while traning models.
# This is not a numerical value so the imputation methods are not available.
movies =  movies[movies['production_companies'] != '[]']

# no movies with original_language of NaN
movies[movies['production_countries'].isna()]
movies =  movies[movies['production_countries'] != '[]']

# movies with release_date of NaN must be deleted in preprocessing,
# imputation methods (average, random value, linear regression) would not
```

```
# work well for the date data type.
# This attribute is actually not a numerical value, so it will be meaingless
# to fill this attribute with the mean, a random value, or a value from linear r
movies = movies[movies['release_date'].notna()]

# no movies with status of NaN
movies[movies['status'].isna()]

# Movies with only released status will be used to train models
# and movies having post production and rumored will be used to test the models.
# This is logically sensible to predict the revenue of unreleased movies, and
# exclude them while training.
movies = movies[movies['status'] == 'Released']
```

In [118…

```
########## tmdb_5000_credits.csv ##########

# This is logically difficult to do imputation on 'cast' attribute
# since this is not a single numerical value.
# movies with 0 cast will be excluded while training.

# no movies with cast of 0 or NaN.
credits[credits['cast'].apply(len) != 0]
credits[credits['cast'].notna()]

# This is logically difficult to do imputation on 'crew' attribute
# since this is not a single numerical value.
# movies with 0 crew will be excluded while training.

# no movies with crew of 0 or NaN.
credits[credits['crew'].apply(len) != 0]
credits[credits['crew'].isna()]
```

Out[118…

movie_id	title	cast	crew

3 Preprocessing (Numeric Values)

Before we started to analyze the numeric data, we eliminated or managed the values below:

(1) zeros, null (NaN), and negative values.

(2) out of range (e.g. ratings more than 10 out of 10).

We regarded these would be inappropriate the anlaysis process.

There were no negative values at all, so two types of actions regarding (1) were made:

- Budget and revenue zeros are deleted from the training set for more accurate model training, but are going to be used in testing set. They are too many just to be eliminated totally from the whole part.
- For vote_average, nothing was changed because we found out that some movies are actually have not been voted or graded even once.

- Other ones are eliminated.

In [119…

```
# Made a new df for checking the values
numeric = ['budget', 'popularity', 'revenue', 'runtime', 'vote_average', 'vote_c
movies[numeric] = movies[numeric].apply(pd.to_numeric, errors='coerce')

numericDf = movies.select_dtypes(include=['number'])

zero = (numericDf == 0).sum()
nan = numericDf.isna().sum()
negative = (numericDf < 0).sum()

countsDf = pd.DataFrame({
    "zeroCount": zero,
    "nanCount": nan,
    "negCount": negative
})

countsDf
```

Out[119…

	zeroCount	nanCount	negCount
budget	759	0	0
id	0	0	0
popularity	0	0	0
revenue	1108	0	0
runtime	5	2	0
vote_average	23	0	0
vote_count	23	0	0

In [120…

```
# Drop NaN / null from numeric values
movies = movies.dropna(subset=numeric)

# Drop Zero values from budget, popularity, revenue, runtime, vote_count
movies = movies[movies["budget"] > 0]
movies = movies[movies["popularity"] > 0]
movies = movies[movies["revenue"] > 0]
movies = movies[movies["runtime"] > 0]
movies = movies[movies["vote_count"] > 0]


# Drop values out of range in terms of voting rate (0 ~ 10)
movies = movies[(movies["vote_average"] >= 0) & (movies["vote_average"] <= 10)]
```

In [121…

```
# Confirm the preprocessing result
zero = (movies == 0).sum()
nan = movies.isna().sum()

newCountDf = pd.DataFrame({
    "zero_count": zero,
    "nan_count": nan
})
```

```
newCountDf
```

Out[121…

	zero_count	nan_count
budget	0	0
genres	0	0
homepage	0	1854
id	0	0
keywords	0	0
original_language	0	0
original_title	0	0
overview	0	0
popularity	0	0
production_companies	0	0
production_countries	0	0
release_date	0	0
revenue	0	0
runtime	0	0
spoken_languages	0	0
status	0	0
tagline	0	223
title	0	0
vote_average	0	0
vote_count	0	0

3.0.1 Create addtional useful coulmns from existing values

The following attributes will be useful for training models.
These convert text or list attributes into numerical values, making non-numerical values suitable for regression models.

1. The number of crews.
2. Average revenue of production countries
3. Average revenue of production companies

The average revenue of each production country

Each movie has one or more production_countries.
Average revenue of each country will be a useful feature for predicting movie

revenue.

In [122…

```
# find the mean revenue of each production country
# https://www.geeksforgeeks.org/python/extract-dictionary-value-from-column-in-d

# expand the data by production countries, and collect the country names, and co
movies['production_countries'] = movies['production_countries'].apply(ast.litera
moviesWithProductionCountry = movies.explode('production_countries')
moviesWithProductionCountry['countryName'] = moviesWithProductionCountry['produc

# find the mean revenue per country, in descending order
revenuePerCountry = moviesWithProductionCountry.groupby('countryName')['revenue'

#find the number of movies of each production country
moviesPerCountry = moviesWithProductionCountry.groupby('countryName')['id'].coun
moviesPerCountry=  moviesPerCountry.to_frame().reset_index().sort_values(by = 'i
moviesPerCountry

# now each movie will have the attribute of mean revenue of each country
moviesWithProductionCountry = moviesWithProductionCountry.merge(revenuePerCountr
moviesWithProductionCountry.rename(
    columns = {
    'revenue_x' : 'revenue',
    'revenue_y' : 'meanRevenuePerCountry'
    },
    inplace = True
)

# for movies that are produced in several countries, find the mean revenue of me
meanRevPerMovie = moviesWithProductionCountry.groupby('id')['meanRevenuePerCount
moviesWithProductionCountry = moviesWithProductionCountry.groupby('id').agg({
    'meanRevenuePerCountry' : 'mean'
}).reset_index()
meanRevPerMovie

# merge the table indicating average revenue, per movie_id with the original dat
movies = movies.merge(moviesWithProductionCountry, on = 'id')

# The top 40 countries with highest revenue
revenuePerCountry.head(40)
```

Out[122…

	countryName	revenue
13	Dominica	1.065660e+09
28	Jamaica	5.626299e+08
33	Malta	4.666267e+08
49	Slovenia	4.196514e+08
4	Bahamas	4.047914e+08
37	New Zealand	3.964483e+08
55	Taiwan	3.390313e+08
41	Philippines	2.032776e+08
11	Czech Republic	1.952173e+08
10	China	1.807956e+08
40	Peru	1.536242e+08
42	Poland	1.445068e+08
59	United Kingdom	1.373491e+08
15	Finland	1.369868e+08
60	United States of America	1.306511e+08
48	Singapore	1.304829e+08
35	Morocco	1.297193e+08
29	Japan	1.296756e+08
2	Australia	1.201023e+08
39	Pakistan	1.172964e+08
58	United Arab Emirates	1.144631e+08
17	Germany	1.086620e+08
7	Bulgaria	1.081282e+08
19	Hong Kong	1.064089e+08
21	Iceland	1.062895e+08
8	Canada	9.386569e+07
20	Hungary	9.219330e+07
22	India	8.548451e+07
27	Italy	8.065999e+07
16	France	7.167086e+07
52	Spain	6.405161e+07
50	South Africa	6.384869e+07
34	Mexico	6.164013e+07

	countryName	revenue
44	Romania	5.853188e+07
43	Portugal	5.840190e+07
53	Sweden	5.818210e+07
45	Russia	4.861752e+07
25	Ireland	4.423427e+07
5	Belgium	4.101746e+07
6	Brazil	4.035071e+07

In [123…

```
# find the mean revenue of each production country
# https://www.geeksforgeeks.org/python/extract-dictionary-value-from-column-in-d

# expand the data by production countries, and collect the country names, and co
movies['production_companies'] = movies['production_companies'].apply(ast.litera
moviesWithProductionCompany = movies.explode('production_companies')
moviesWithProductionCompany['companyName'] = moviesWithProductionCompany['produc

# find the mean revenue per country, in ascending order
revenuePerCompany = moviesWithProductionCompany.groupby('companyName')['revenue'

#find the number of movies of each production country
moviesPerCompany = moviesWithProductionCompany.groupby('companyName')['id'].coun
moviesPerCompany=  moviesPerCompany.to_frame().reset_index().sort_values(by = 'i
moviesPerCompany

# now each movie will have the attribute of mean revenue of each country
moviesWithProductionCompany = moviesWithProductionCompany.merge(revenuePerCompan
moviesWithProductionCompany.rename(
    columns = {
    'revenue_x' : 'revenue',
    'revenue_y' : 'meanRevenuePerCompany'
    },
    inplace = True
)

# for movies that are produced in several countries, find the mean revenue of me
meanRevPerMovie = moviesWithProductionCompany.groupby('id')['meanRevenuePerCompa
moviesWithProductionCompany = moviesWithProductionCompany.groupby('id').agg({
    'meanRevenuePerCompany' : 'mean'
}).reset_index()
meanRevPerMovie

# merge the table indicating average revenue, per movie_id with the original dat
movies = movies.merge(moviesWithProductionCompany, on = 'id')

# The top 40 companies with highest revenue
revenuePerCompany.head(40)
```

Out[123…

	companyName	revenue
675	Colorado Office of Film, Television & Media	1.506249e+09
74	Abu Dhabi Film Commission	1.506249e+09
2546	Québec Production Services Tax Credit	1.506249e+09
2493	Prime Focus	1.405404e+09
3381	Vita-Ray Dutch Productions (III)	1.153304e+09
831	Deluxe Digital Studios	1.153304e+09
1802	Lightstorm Entertainment	1.124376e+09
1467	Ian Bryce Productions	1.091405e+09
2777	Second Mate Productions	1.013330e+09
1033	Fairview Entertainment	9.665506e+08
29	3Foot7	9.560198e+08
3156	The Saul Zaentz Company	8.988279e+08
2384	Patalex IV Productions Limited	8.959210e+08
590	China Movie Channel	8.868676e+08
261	B24	8.806746e+08
744	Cruel & Unusual Films	8.732602e+08
1999	Media Rights Capital (MRC)	8.616360e+08
3124	The Donners' Company	7.831130e+08
1966	Marvel Studios	7.798770e+08
460	Bulletproof Cupid	7.733286e+08
1373	Hasbro Studios	7.729499e+08
1046	Farewell Productions	7.696536e+08
1504	Indochina Productions	7.656750e+08
1477	Illumination Entertainment	7.461838e+08
1385	Heineken Branded Entertainment	7.385997e+08
1204	Fuji Television Network	7.259420e+08
587	China Film Co.	7.138663e+08
2932	SprocketHeads	7.097098e+08
1976	Matt Tolmach Productions	7.057174e+08
255	Avi Arad Productions	7.057174e+08
2263	Odin	6.823301e+08
1687	Kinberg Genre	6.635239e+08
3482	WingNut Films	6.437639e+08

	companyName	revenue
1396	Heyday films	6.308280e+08
387	Blue Light	6.240294e+08
1220	GH Three	6.240294e+08
1395	Heyday Films	6.217698e+08
2445	Pixar Animation Studios	6.131120e+08
1546	Internationale Filmproduktion Richter	6.098416e+08
1824	Littlestar	6.098416e+08

In [124…

```
# now each movies have the mean revenue of countries
movies
```

Out[124...

	budget	genres	homepage	id	
0	237000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.avatarmovie.com/	19995	
1	300000000	[{"id": 12, "name": "Adventure"}, {"id": 14, "...	http://disney.go.com/disneypictures/pirates/	285	[{"i "oc
2	245000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.sonypictures.com/movies/spectre/	206647	[{"i "
3	250000000	[{"id": 28, "name": "Action"}, {"id": 80, "nam...	http://www.thedarkknightrises.com/	49026	[{"i "d
4	260000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://movies.disney.com/john-carter	49529	[{"i "b
...	...	...	...	...	
3178	27000	[{"id": 35, "name": "Comedy"}]	http://www.miramax.com/movie/clerks/	2292	"s
3179	12000	[{"id": 27, "name": "Horror"}, {"id": 35, "nam...	NaN	692	[{"i "
3180	20000	[{"id": 80, "name": "Crime"}, {"id": 27, "name...	NaN	36095	[{"i "ja
3181	7000	[{"id": 878, "name": "Science Fiction"}, {"id"...	http://www.primermovie.com	14337	"na

	budget	genres	homepage	id	
3182	220000	[{"id": 28, "name": "Action"}, {"id": 80, "nam...	NaN	9367	stat

3183 rows × 22 columns

The number of crews for each movie

It is a reasonable assumpiton that the number of crews who invlolved in producing a movie is related to the quality of the movie, which may result in higher revenue. Therefore, it is valuable to examine how the number of crews affects revenue.

In [125…

```
# Crew
# The number of crews for each movie

# convert string to dictionary
credits['crew'] = credits['crew'].apply(ast.literal_eval)
credits['numberOfCrews'] = credits['crew'].str.len()
credits
```

Out[125…

	movie_id	title	cast	crew	numberOfCrew
0	19995	Avatar	[{"cast_id": 242, "character": "Jake Sully", "...	[{'credit_id': '52fe48009251416c750aca23', 'de...	153
1	285	Pirates of the Caribbean: At World's End	[{"cast_id": 4, "character": "Captain Jack Spa...	[{'credit_id': '52fe4232c3a36847f800b579', 'de...	32
2	206647	Spectre	[{"cast_id": 1, "character": "James Bond", "cr...	[{'credit_id': '54805967c3a36829b5002c41', 'de...	155
3	49026	The Dark Knight Rises	[{"cast_id": 2, "character": "Bruce Wayne / Ba...	[{'credit_id': '52fe4781c3a36847f81398c3', 'de...	217
4	49529	John Carter	[{"cast_id": 5, "character": "John Carter", "c...	[{'credit_id': '52fe479ac3a36847f813eaa3', 'de...	132
...	...	...	...	...	.
4798	9367	El Mariachi	[{"cast_id": 1, "character": "El Mariachi", "c...	[{'credit_id': '52fe44eec3a36847f80b280b', 'de...	11
4799	72766	Newlyweds	[{"cast_id": 1, "character": "Buzzy", "credit_...	[{'credit_id': '52fe487dc3a368484e0fb013', 'de...	8
4800	231617	Signed, Sealed, Delivered	[{"cast_id": 8, "character": "Oliver O₩u2019To...	[{'credit_id': '52fe4df3c3a36847f8275ecf', 'de...	13
4801	126186	Shanghai Calling	[{"cast_id": 3, "character": "Sam", "credit_id...	[{'credit_id': '52fe4ad9c3a368484e16a36b', 'de...	2
4802	25975	My Date with Drew	[{"cast_id": 3, "character": "Herself", "credi...	[{'credit_id': '58ce021b9251415a390165d9', 'de...	5

4803 rows × 5 columns

In [126…

```
#merge two csv files into one
```

```
credits.drop(columns = 'title', inplace = True)
movies = movies.merge(credits, left_on = 'id', right_on = 'movie_id', how = 'lef
movies = movies.drop(columns = ['movie_id'])
```

3.0.2 Preprocessing Done.

At this point, we have finished the preprcessing stage of given data files, tmdb_5000_movies.csv and tmdb_5000_credits.csv.
Since data in two files have been unified into one data frame,
it becomes more conveinient to generate statistics and training models.
Additionally, we have added usesful columns such as the numebr of crews, and the average revenue per a production country, which is to help the models learn bettwe pattterns and produce more accurate predictions on revenue.

4 Data Analysis

4.1.1 (1) Movies released by time periods

Count the number of movies released by day of week, month and year. Are there any patterns that you observe?

In [127…

```
# group movies by data (week)

#convert the data type of 'release_date' from Object to DateTime.
movies['release_date'] = pd.to_datetime(movies['release_date'])

# copy the original movies data
moviesWithDateTime = movies.copy()

# Extract specific day, month, and year from 'release_date'
moviesWithDateTime['day'] = movies['release_date'].dt.day_name()
moviesWithDateTime['month'] = movies['release_date'].dt.month_name()
moviesWithDateTime['year'] = movies['release_date'].dt.year
```

4.1.1 The number of movies by a day of the week.

Movies tend to be released more frequently as the week approaches Friday.
Movies are released the most on Friday,
whille the fewest movies are released on the weekend.

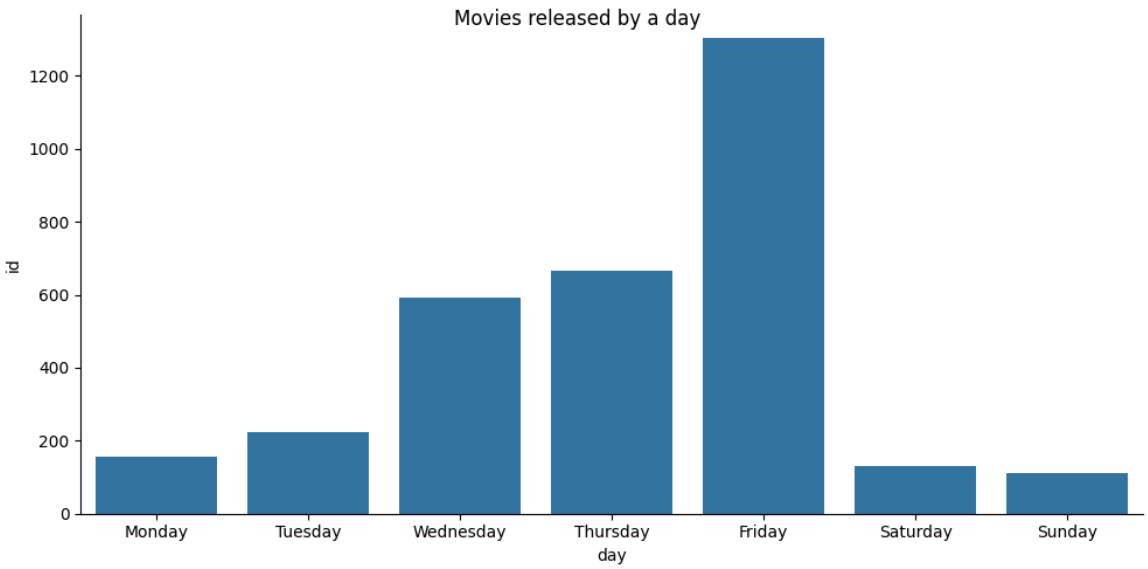
In [128…

```
# The number of movies by a day of the week.
order = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Su

byADay = sns.catplot(
    data = moviesWithDateTime.groupby('day')['id'].count().to_frame().reset_inde
    x = 'day',
    y = 'id',
    kind = 'bar',
    order = order,
    aspect = 2
```

```
)

byADay.figure.suptitle('Movies released by a day')

moviesWithDateTime.groupby('day')['id'].count().to_frame().reset_index()
```

Out[128…

	day	id
0	Friday	1303
1	Monday	157
2	Saturday	129
3	Sunday	112
4	Thursday	666
5	Tuesday	223
6	Wednesday	593



4.1.2 The number of movies released by a month

Movies are released the most on Sebtember,
and the fewest movies are released on November.

Movies tend to be released more frequently as the year moves toward its end.

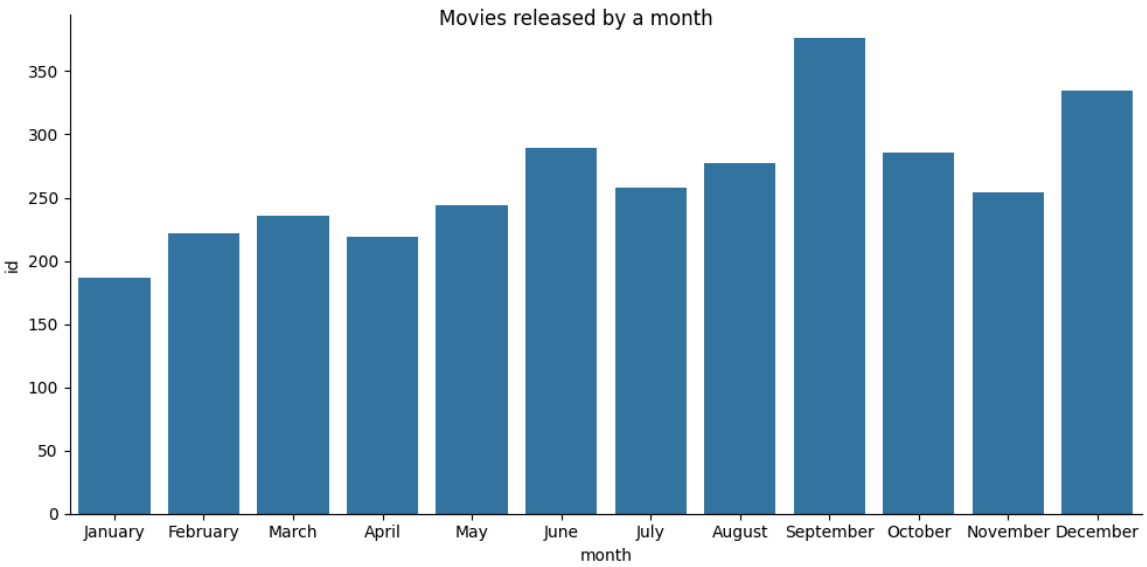
In [129…

```
# The number of movies released by a month

order = [
    'January',
    'February',
    'March',
    'April',
    'May',
    'June',
    'July',
    'August',
    'September',
    'October',
```

```
    'November',
    'December'
]

byAMonth = sns.catplot(
    data = moviesWithDateTime.groupby('month')['id'].count().to_frame().reset_in
    x = 'month',
    y = 'id',
    kind = 'bar',
    order = order,
    aspect = 2
)
byAMonth.figure.suptitle('Movies released by a month')

moviesWithDateTime.groupby('month')['id'].count().to_frame().reset_index()
```

Out[129…

	month	id
0	April	219
1	August	277
2	December	335
3	February	222
4	January	187
5	July	258
6	June	289
7	March	236
8	May	244
9	November	254
10	October	286
11	September	376



4.1.3 The number of movies released by a year

As the time passes, the number of movies being released increases steadily.
In the graph below, movies releases grow almost exponentially.

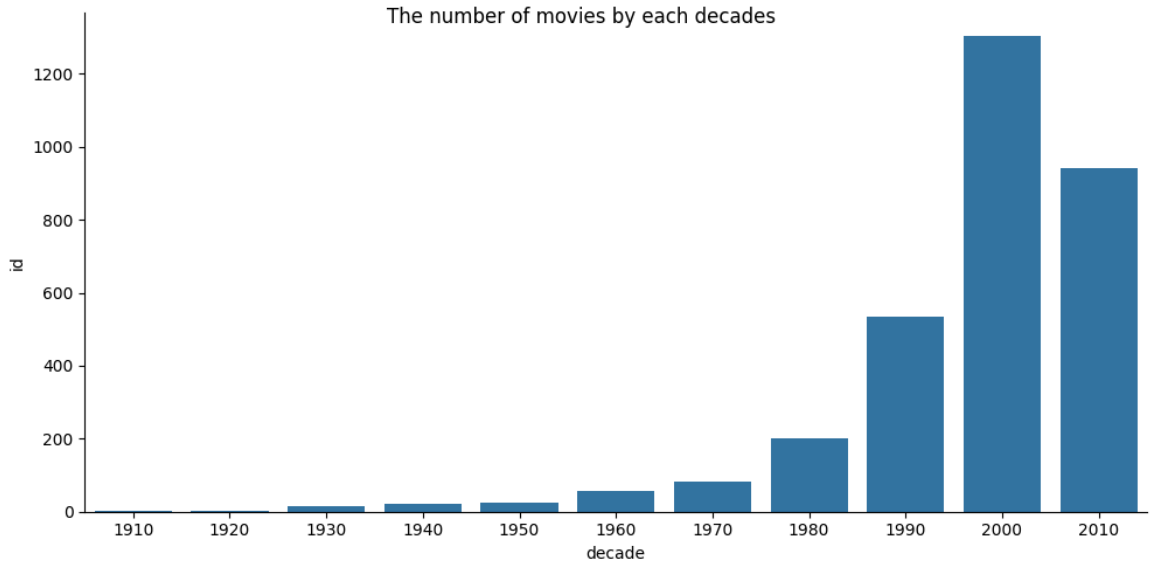
The lastet movie in the data was released in 2017,
so the number of movies that are released in 2010s is likely lower than
the actual total number because the data does not include the full decade.

In [130…

```
# The number of movies released by a year

# The 'decade' column is used to group movies into 10-year intervals.
moviesWithDateTime['decade'] = (moviesWithDateTime['year'] // 10) * 10

byAMonth = sns.catplot(
    data = moviesWithDateTime.groupby('decade')['id'].count().to_frame().reset_i
    x = 'decade',
    y = 'id',
    kind = 'bar',
    aspect = 2
)
byAMonth.figure.suptitle('The number of movies by each decades')

moviesWithDateTime.groupby('decade')['id'].count().to_frame().reset_index()
```

Out[130…

	decade	id
0	1910	1
1	1920	3
2	1930	14
3	1940	20
4	1950	25
5	1960	58
6	1970	81
7	1980	202
8	1990	535
9	2000	1303
10	2010	941



4.2 (2) Genre trend over time

We tried to analyze the movie genre trend over decades, by sorting how many movies of particular genre are made.

For conciseness, a bar graph has used to indicate the top 3 genres that had made the most.

We had several insights, but the most obvious point was that the movie market was the **most active in 2000's** than any other decades. Also, **Drama was one of the most popular** genres across all time periods.

In [131…

```
# Extracted release year from release date column
movies["release_date"] = pd.to_datetime(movies["release_date"], errors="coerce")
movies["release_year"] = movies["release_date"].dt.year

movies[["release_date", "release_year"]].head()
```

Out[131…

	release_date	release_year
0	2009-12-10	2009
1	2007-05-19	2007
2	2015-10-26	2015
3	2012-07-16	2012
4	2012-03-07	2012

In [132…

```
# made a distinct genre list
genreList = []
for val in movies["genres"]:
    if pd.isna(val):
        genreList.append([])
        continue

    try:
```

```
        parsed = ast.literal_eval(val)
    except (ValueError, SyntaxError):
        genreList.append([])
        continue

    if isinstance(parsed, list):
        genreList.append(
            [d.get("name") for d in parsed if isinstance(d, dict)]
        )
    else:
        genreList.append([])

movies["genre_names"] = genreList
genreDf = movies.explode("genre_names").rename(columns={"genre_names": "genre"})

# changed release year to decades
genreDf = genreDf.dropna(subset=["release_year"])
genreDf["decade"] = (genreDf["release_year"] // 10) * 10

# genre counts for each decades
genreDecadeCounts = (
    genreDf
    .groupby(["decade", "genre"])
    .size()
    .reset_index(name="count")
)

# Extracted only top 3 among them and made a df
top3 = (
    genreDecadeCounts
    .sort_values(["decade", "count"], ascending=[True, False])
    .groupby("decade")
    .head(3)
    .copy()
)

top3["rank"] = top3.groupby("decade")["count"].cumcount() + 1

top3
```

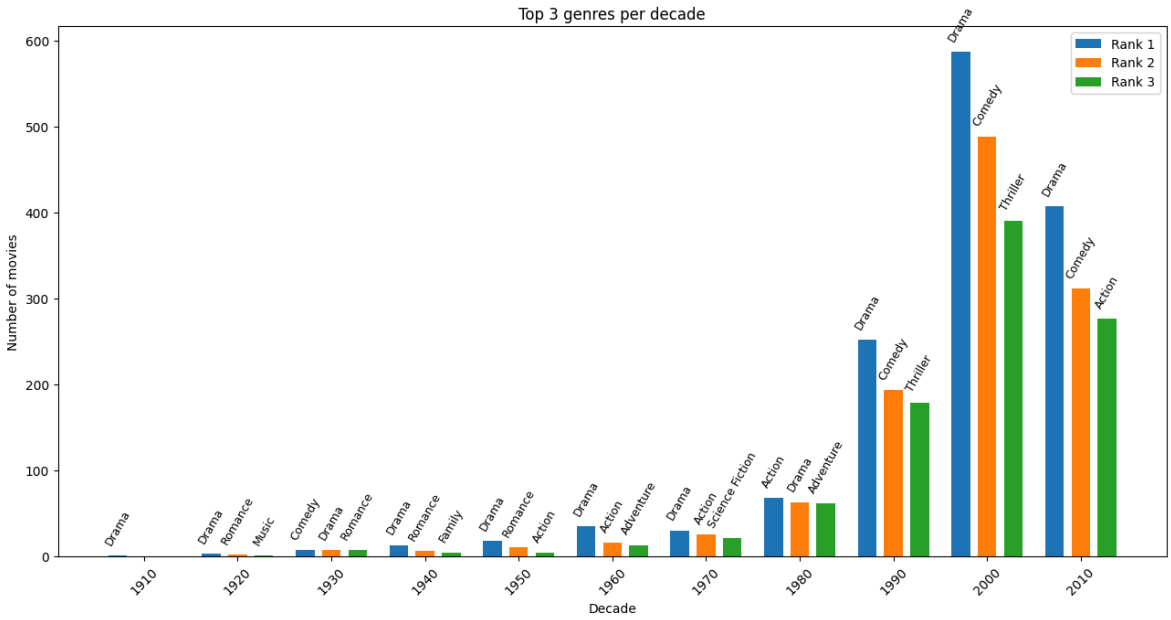
Out[132…

	decade	genre	count	rank
0	1910	Drama	1	1
1	1920	Drama	3	1
3	1920	Romance	2	2
2	1920	Music	1	3
9	1930	Comedy	7	1
10	1930	Drama	7	2
15	1930	Romance	7	3
21	1940	Drama	13	1
27	1940	Romance	6	2
22	1940	Family	4	3
35	1950	Drama	18	1
40	1950	Romance	10	2
31	1950	Action	4	3
49	1960	Drama	35	1
45	1960	Action	16	2
46	1960	Adventure	13	3
66	1970	Drama	30	1
61	1970	Action	25	2
74	1970	Science Fiction	21	3
78	1980	Action	68	1
84	1980	Drama	63	2
79	1980	Adventure	61	3
102	1990	Drama	252	1
99	1990	Comedy	193	2
111	1990	Thriller	179	3
120	2000	Drama	587	1
117	2000	Comedy	488	2
130	2000	Thriller	390	3
139	2010	Drama	407	1
136	2010	Comedy	311	2
133	2010	Action	276	3

In [133…

```
# bar plot
barPositions = {}
```

```
decades = sorted(top3["decade"].unique())
ranks = [1, 2, 3]

pivot = top3.pivot(index="decade", columns="rank", values="count").reindex(decad
x = np.arange(len(decades))

barWidth = 0.2
innerGap = 0.08
groupWidth = len(ranks) * barWidth + (len(ranks) - 1) * innerGap

plt.figure(figsize=(13, 7))

for i, r in enumerate(ranks):
    heights = pivot[r].values

    positions = (
        x - groupWidth / 2
        + barWidth / 2
        + i * (barWidth + innerGap)
    )


    barPositions[r] = positions

    plt.bar(positions, heights, width = barWidth, label = f"Rank {r}")

plt.xticks(x, decades, rotation = 45)
plt.xlabel("Decade")
plt.ylabel("Number of movies")
plt.title("Top 3 genres per decade")

plt.legend()

# text positions
for i, decade in enumerate(decades):
    for j, r in enumerate(ranks):
        row = top3[(top3["decade"] == decade) & (top3["rank"] == r)]
        if not row.empty:
            genreName = row["genre"].iloc[0]
            count = row["count"].iloc[0]
            xpos = barPositions[r][i]
            ypos = count + 10
            plt.text(xpos, ypos, genreName, ha="center", va="bottom", rotation=6

plt.tight_layout()
plt.show()
```



4.3 (3) Budget vs. Revenue

Above two analysis were based on the questions of the instruction, but now we would like to find another interesting insight.

This analysis will focus on calculating and ranking movies by their Return on Investment (Revenue/Budget), excluding all non-commercial zero-value entries. The process involves creating a new ROI ratio column, followed by sorting the dataset to identify the most financially efficient productions.

In ranking Top 10, there were data errors as we learned from the lecture slides. When you see the budget column, there are some values that seems to be nonsense, such as **1, 10, 218**.

We searched those movie budgets on google by our own, and found out that they were `$` **1.5M,** `$` **10M, and** `$` **218** each. Surprising thing was that the movie **Tarnation** was actually one of the famous low-budget made movies.

However, the **Modern Times** and **Nurse 3-D** can concluded that this error likely originated from negligence on the part of the data entry personnel or the platform itself. The correct values should have been entered with consideration for the units, or the platform should have properly conducted unit conversion during the process of data refinement and management.

So if the first two are excluded from the ranking, **Paranormal Activity** would be the movie having highest ROI.

In [134…

```
moviesRoi = movies[(movies['budget'] > 0) & (movies['revenue'] > 0)].copy()

moviesRoi['ROI_ratio'] = moviesRoi['revenue'] / moviesRoi['budget']

moviesRoi['genres'] = moviesRoi['genres'].apply(lambda x: ast.literal_eval(x) if
```

```
bestRoiMovies = moviesRoi.sort_values(by = 'ROI_ratio', ascending = False).head(

print(bestRoiMovies[['title', 'release_year', 'genre_names', 'budget', 'revenue'
```

```
                           title  release_year  \
3046                 Modern Times          1936
2513                    Nurse 3-D          2013
3131           Paranormal Activity          2007
3116       The Blair Witch Project          1999
3173                    Eraserhead          1977
3179                Pink Flamingos          1972
3175                 Super Size Me          2004
3172                   The Gallows          2015
3120                    Open Water          2004
2524  The Texas Chain Saw Massacre          1974

                                  genre_names  budget    revenue  \
3046                           [Drama, Comedy]       1    8500000
2513                        [Horror, Thriller]      10   10000000
3131                         [Horror, Mystery]   15000  193355800
3116                         [Horror, Mystery]   60000  248000000
3173  [Drama, Fantasy, Horror, Science Fiction]   10000    7000000
3179                    [Horror, Comedy, Crime]   12000    6000000
3175               [Documentary, Comedy, Drama]   65000   28575078
3172                        [Horror, Thriller]  100000   42664410
3120                         [Drama, Thriller]  130000   54667954
2524                                   [Horror]   85000   30859000

         ROI_ratio
3046  8.500000e+06
2513  1.000000e+06
3131  1.289039e+04
3116  4.133333e+03
3173  7.000000e+02
3179  5.000000e+02
3175  4.396166e+02
3172  4.266441e+02
3120  4.205227e+02
2524  3.630471e+02
```

4.4 (4) Correlation Between Features

This heatmap, including both numeric and non-numeric features below reveals that which features strongly or weakly affect movie revenue.

Features having strong correlations with revenue:

- popularity
- vote_count
- budget
- meanRevenuePerCountry

Features having weak correlations with revenue:

- numberOfCrews
- meanRevenuePerCountry

- vote_average

It is noticeable that features having the lowest correlation with revenue is vote_average, so this will not be used while traning models.

In [135…

```
correlations = movies[['revenue',
                       'numberOfCrews',
                       'popularity',
                       'vote_count',
                       'meanRevenuePerCountry',
                       'budget',
                       'vote_average',
                       'meanRevenuePerCompany'
                       ]].corr()
correlations

# create a heatmap
sns.heatmap(correlations, annot = True)
```

Out[135…

```
<Axes: >
```

5 Build Models

Three models will be used:

- Simple linear regression (baseline model)

- Multiple linear regression
- Polynomial regression

For cross-validation, data will be split based on:

- movies released before vs. after 2010
- random sampling

How each model work?

Simple linear regression

- Evaluate a straight line to the data. The model assumes the relationship between x and y is linear and finds the optimistic parameters a and b in y = ax + b

by minimizing the least squares error between predicted and actual values, as taught in the lecture.

Multiple linear regression

- Although the model is still linear, it uses multiple input variables unlike simple linear regression.
- Each variable has its own independent coefficient (weight).
- The form of equation will be: y = a_0 + a_0x_0 + a_1x_1 + ... + a_nx_n
- Using methods such as least squares, the model calculates the coefficients and the intercept that minimize the total prediction error.

Polynomial regression

- Now the relationship between inputs and a target does not guarantee a linearity.
- When multiple input variables exist, then polynomial regression considers combinations of input variables

(a multiplication of them for example; x_1 * x_2) to capture complex patterns and improve model performance.

Feature Variables used in training

- Log Budget
- Popularity
- Vote count
- Number of Crews
- Log mean revenue per Country
- Log Mean revenue per production company

Variables used for visualization of results

- Log Budget (x-axis)
- Log Revenue (y-axis)

Budget was used as x-axis in the result graphs because the RSE difference between models using budget and vote_count was minimal (around 0.0xx), suggesting similar predictive performace.

Although vote_count has a higher correlation, budget still works well as it captures multiple underlyung aspects of a film's production.

Log transformation was applied to both revenue and budget for scale reduction.

5.1 Case 1: Split data for training and testing models (Before vs. after 2010)

Training set:
Data before 2010's (~ 2009)

Testing set:
Data of 2010's (2010 ~)

Targeting:
Prediction of every revenue in 2010's

This time-based split is resonable because it simulates a real-world scenario **(using past data to predict future outcomes)**.

In [136…

```
# Extract data which will be used for training
trainingMovies = movies[movies['release_date'].dt.year < 2010].copy()
testingMovies = movies[movies['release_date'].dt.year >= 2010].copy()
```

5.1.1 Train Baseline Model - Linear Regression

The following model will be a baseline model for predicting movie revenue by using only one feature, budgets.

Equation:

- y = 0.7643952053191282 * x + 4.667164913993144

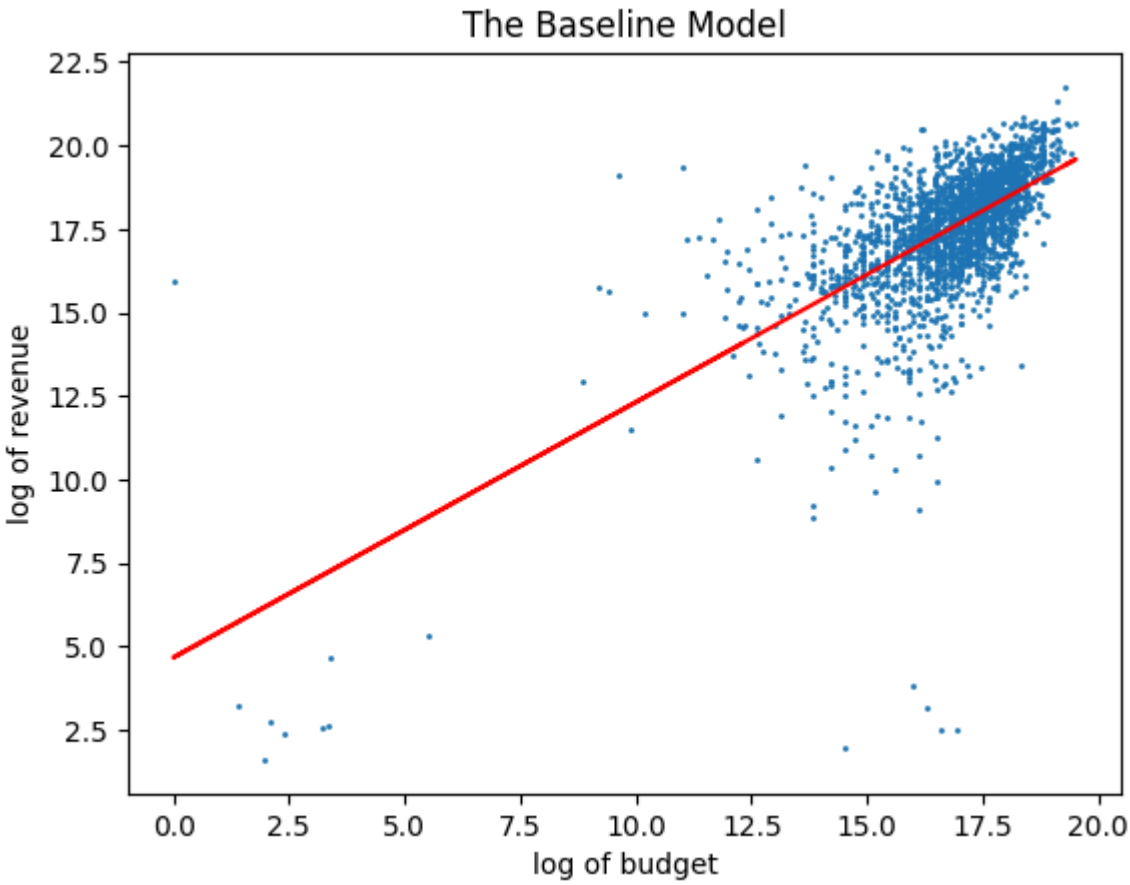
RSE using training data : 1.5531

In [137…

```
from scipy import stats
# create a base line model using linear regression from scipy stats.
# https://www.w3schools.com/python/python_ml_linear_regression.asp

# use log to normalize data
trainingMovies['logBudget'] = np.log(trainingMovies['budget'])
trainingMovies['logRevenue'] = np.log(trainingMovies['revenue'])

x = trainingMovies['logBudget']
y = trainingMovies['logRevenue']

slope, intercept, r, p, std_err = stats.linregress(x, y)
```

```
def myfunc(x):
    return slope * x + intercept

# linear regression equation using scipy stats
print("Equation: ")
print("y = ", str(slope), "* x +", str(intercept))
mymodel = list(map(myfunc, x))

# The data graph including the linear regression function
plt.scatter(x, y, s = 1)
plt.plot(x, mymodel, color = "red")
plt.xlabel("log of budget")
plt.ylabel("log of revenue")
plt.title("The Baseline Model")
plt.show()

# Calculate RSE using traningMovies
prediectedY = myfunc(x)
n = len(y)
RSE = np.sqrt(np.sum((y - prediectedY)**2) / (n - 2))
print("RSE:", RSE)
```

```
Equation:
y =  0.7643952053191282 * x + 4.667164913993144
```




```
RSE: 1.5531147381099775
```

5.1.1 Test the baseline model

Since the equation of the baseline model has been computed, it is possible to evaluate how model is accurate by comparing the actual revenues with the revenues predicted by the model.

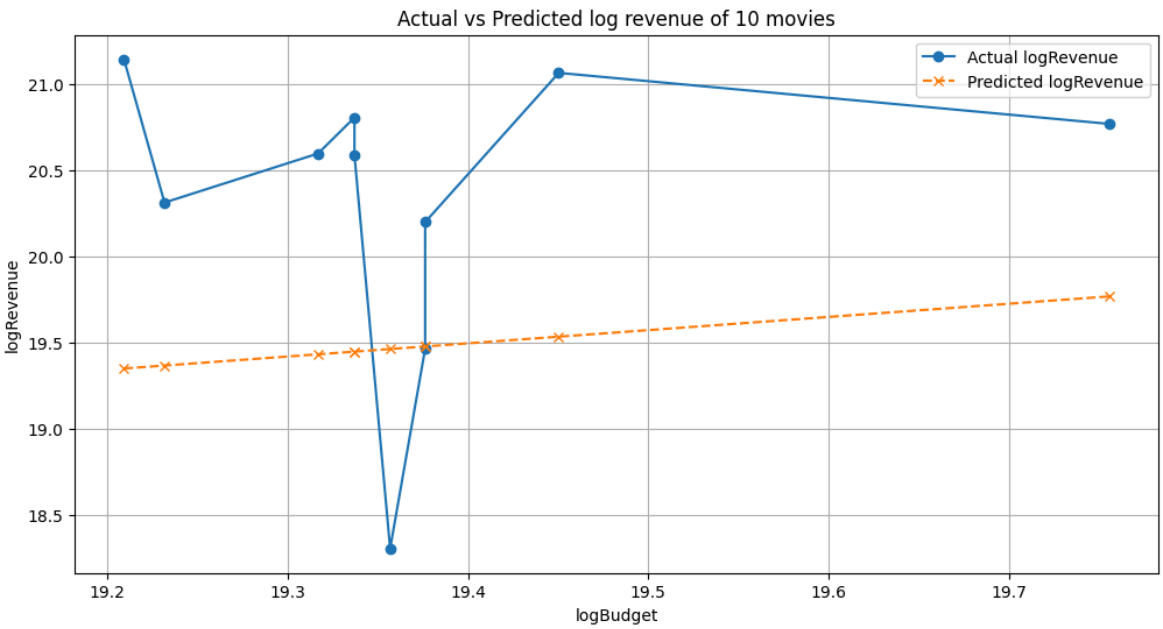
RSE using testing data: 1.5213

Since the model follows a linear form y = ax + b, it is visually clear that the predicted values differ noticeably from the actual values.
Revenue cannot be predicted accurately by only a single feature like budget.

In [138…

```
# movies released after 2010.
testingMovies

# use log to normalize datas
testingMovies['logBudget'] = np.log(testingMovies['budget'])
testingMovies['logRevenue'] = np.log(testingMovies['revenue'])


y_tested = myfunc(testingMovies['logBudget'])
y_tested
testingMovies['logRevenue']

# RSE using testing data
n = len(y_tested)
RSE = np.sqrt(np.sum((y_tested - testingMovies['logRevenue'])**2) / (n - 2))
print("RSE: ", RSE)

testingMovies['predicted_logRevenue'] = y_tested
top10Movies = testingMovies.head(10)
top10Movies = top10Movies.sort_values(by = 'logBudget')

plt.figure(figsize=(12, 6))
plt.plot(top10Movies['logBudget'], top10Movies['logRevenue'], label = 'Actual lo
plt.plot(top10Movies['logBudget'], top10Movies['predicted_logRevenue'], label =
plt.title('Actual vs Predicted log revenue of 10 movies')
plt.xlabel('logBudget')
plt.ylabel('logRevenue')
plt.legend()
plt.grid(True)
plt.show()
```

```
RSE:  1.5213008203615246
```



5.1.2 Train Multiple Linear Regression model

Based on the previous data analysis, the heatmap showed that movie revenue is correlated not only with budget but also with several other features.
Therefore, the multiple linear regression model will be used to estimate the revenue more accrurately, using multiple features.

RSE using testing data: 1.1953

In [139…

```
from sklearn import linear_model

# take log of meanRevenuePerCountry to normalize the data
trainingMovies['logMeanRevenuePerCountry'] = np.log(trainingMovies['meanRevenueP
trainingMovies['logMeanRevenuePerCompany'] = np.log(trainingMovies['meanRevenueP

testingFeatures = ['logBudget', 'popularity', 'vote_count', 'numberOfCrews', 'lo
x = trainingMovies[testingFeatures]
y = trainingMovies['logRevenue']

regr = linear_model.LinearRegression()
regr.fit(x, y)

print("Weights (w_i):", regr.coef_)
print("Intercept (w_0):", regr.intercept_)

y_pred = regr.predict(x)
RSE = np.sqrt(np.sum((y - y_pred)**2) / (len(y) - 6 - 1))
print("RSE: ", RSE)
```

```
Weights (w_i): [4.53440337e-01 2.04840487e-02 2.16889044e-05 7.84954029e-04
 3.79815925e-03 6.63771875e-01]
Intercept (w_0): -2.866255162514438
RSE:  1.1952733098056751
```

5.1.2 Test the multiple regression model

Since the multiple regression model has been trained, it is possible to evaluate how model is accurate by comparing the actual revenues with the revenues predicted by the model.

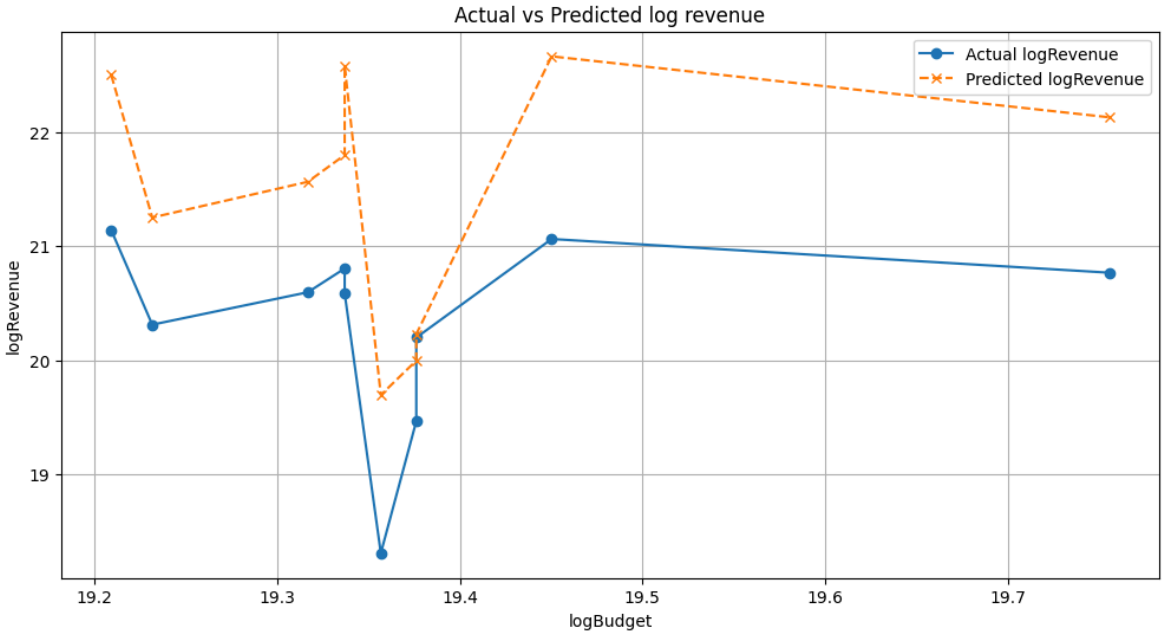
RSE using training data: 1.4588

The RSE of the testing data is higher than that of the training data.
This indicates that the movie industry changed after 2010, meaning that a model trained only on movies released before 2010 cannot fully capture the revenue patterns of movies released after 2010.

In [140…

```
# movies released after 2010.
testingMovies

# take log of meanRevenuePerCountry to normalize the data
testingMovies['logMeanRevenuePerCountry'] = np.log(testingMovies['meanRevenuePer
testingMovies['logMeanRevenuePerCompany'] = np.log(testingMovies['meanRevenuePer
```

```
testingY_pred = regr.predict(testingMovies[testingFeatures])

# RSE using testing data
n = len(testingY_pred)
RSE = np.sqrt(np.sum((testingY_pred - testingMovies['logRevenue'])**2) / (n - 6
RSE

testingMovies['predicted_logRevenue'] = testingY_pred
top10Movies = testingMovies.head(10)
top10Movies = top10Movies.sort_values(by = 'logBudget')

plt.figure(figsize=(12, 6))
plt.plot(top10Movies['logBudget'], top10Movies['logRevenue'], label = 'Actual lo
plt.plot(top10Movies['logBudget'], top10Movies['predicted_logRevenue'], label =
plt.title('Actual vs Predicted log revenue')
plt.xlabel('logBudget')
plt.ylabel('logRevenue')
plt.legend()
plt.grid(True)
plt.show()
```



5.1.3 Train Polynomial Regression model

Since we have already estimated movie revenues using a multiple linear regression model with several features, we now investigate whether a polynomial relationship between revenue and features produces better results.
Therefore, a polynomial regression model will be employed to estimate the revenue more accurately using the same set of features.

RSE using training data when degree = 2 and alpha = 1

- 1.0803

Degree of 2 and alpha of 1 were selected because these provide the best balance between complexity and stability of the polynomial model (Optimistic values for bias-variance tradeoffs).

In [141…

```
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import Ridge
from sklearn.preprocessing import StandardScaler

x = trainingMovies[['logBudget', 'popularity', 'vote_count', 'numberOfCrews', 'l
y = trainingMovies['logRevenue']

poly = PolynomialFeatures(degree = 2, include_bias = False)

# create feature matrix
scaler = StandardScaler()
scaledX = scaler.fit_transform(x)
x = poly.fit_transform(scaledX)

# train models
model = Ridge(alpha = 1)
model.fit(x, y)

# collect predicted results
y_predicted = model.predict(x)

result = pd.DataFrame({
    'logBudget' : trainingMovies['logBudget'],
    'y_predicted' : y_predicted
})

# residual sum of squares (RSS)
n = len(y)
RSE = np.sqrt(np.sum((y - y_predicted) ** 2) / (n - 6 - 1))
print("RSE:", RSE)
```

```
RSE: 1.0803132909493016
```

5.1.3 Test the polynomial regression model

Since the polynomial regression model has been trained, it is possible to evaluate how model is accurate by comparing the actual revenues with the revenues predicted by the model.

RSE using training data: 1.9158

The RSE of the testing data turned out to be higher than the RSE of the training data.

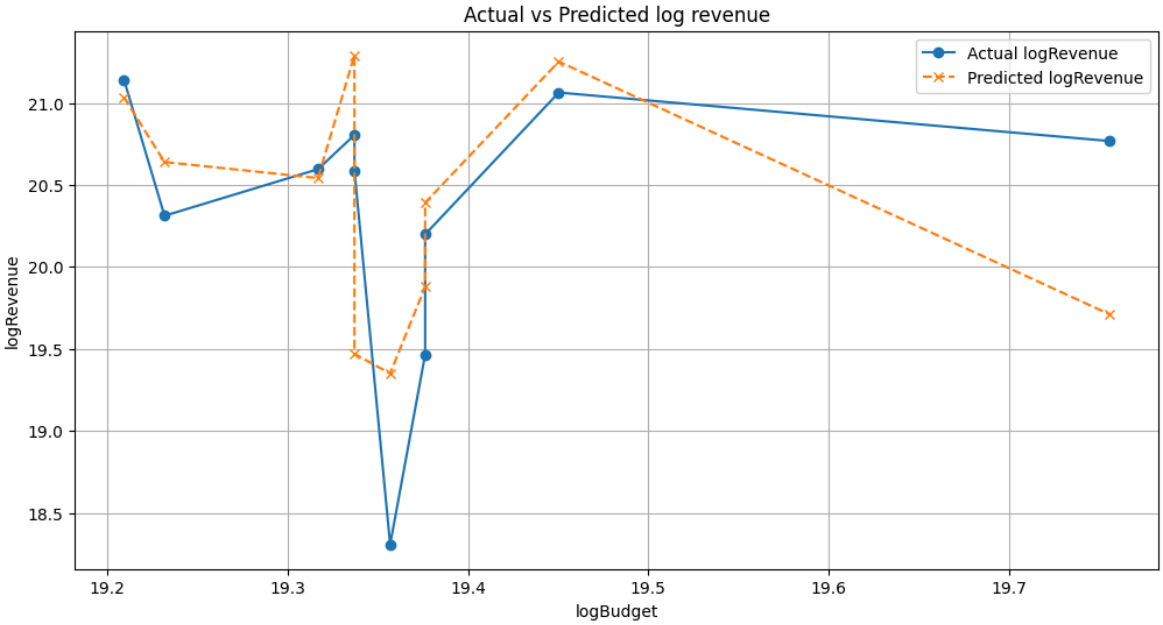
Similar to the results from multiple linear regression, the model does not capture the changed tendency of movies released after 2010.
The gap between the RSE of training and testing data becomes larger when compared to the RSEs from the multiple linear regression model.
This model is more complex, which increases the risk of overfitting, resulting in high variance on the testing data.

In [142…

```
# movies released after 2010.
testingMovies

x = testingMovies[['logBudget', 'popularity', 'vote_count', 'numberOfCrews', 'lo
y = testingMovies['logRevenue']
```

```
scaledX = scaler.transform(x)
x = poly.transform(scaledX)

testingY_pred = model.predict(x)
# print(testingY_pred)

# RSE using testing data
n = len(testingY_pred)
RSE = np.sqrt(np.sum((y - testingY_pred)**2) / (n - 6 - 2))
print("RSE: ", RSE)

testingMovies['predicted_logRevenue'] = testingY_pred
top10Movies = testingMovies.head(10)
top10Movies = top10Movies.sort_values(by = 'logBudget')

plt.figure(figsize=(12, 6))
plt.plot(top10Movies['logBudget'], top10Movies['logRevenue'], label='Actual logR
plt.plot(top10Movies['logBudget'], top10Movies['predicted_logRevenue'], label='P
plt.title('Actual vs Predicted log revenue')
plt.xlabel('logBudget')
plt.ylabel('logRevenue')
plt.legend()
plt.grid(True)
plt.show()
```

```
RSE:  1.9157873105581031
```



5.2 Case 2: Split data for training and testing models (Random Sampling)

Training set:

Randomly selected 70% from whole data

Testing set:

Rest 30% of whole data

Targeting:
Prediction of every revenue in randomly selected data

The movies used to train and test the models will be selected randomly.

In [143…
```
from sklearn.model_selection import train_test_split

# movie data is randomly split
trainingMovies, testingMovies = train_test_split(movies, test_size = 0.3, random
```

5.2.1 Train Baseline Model - Linear Regression

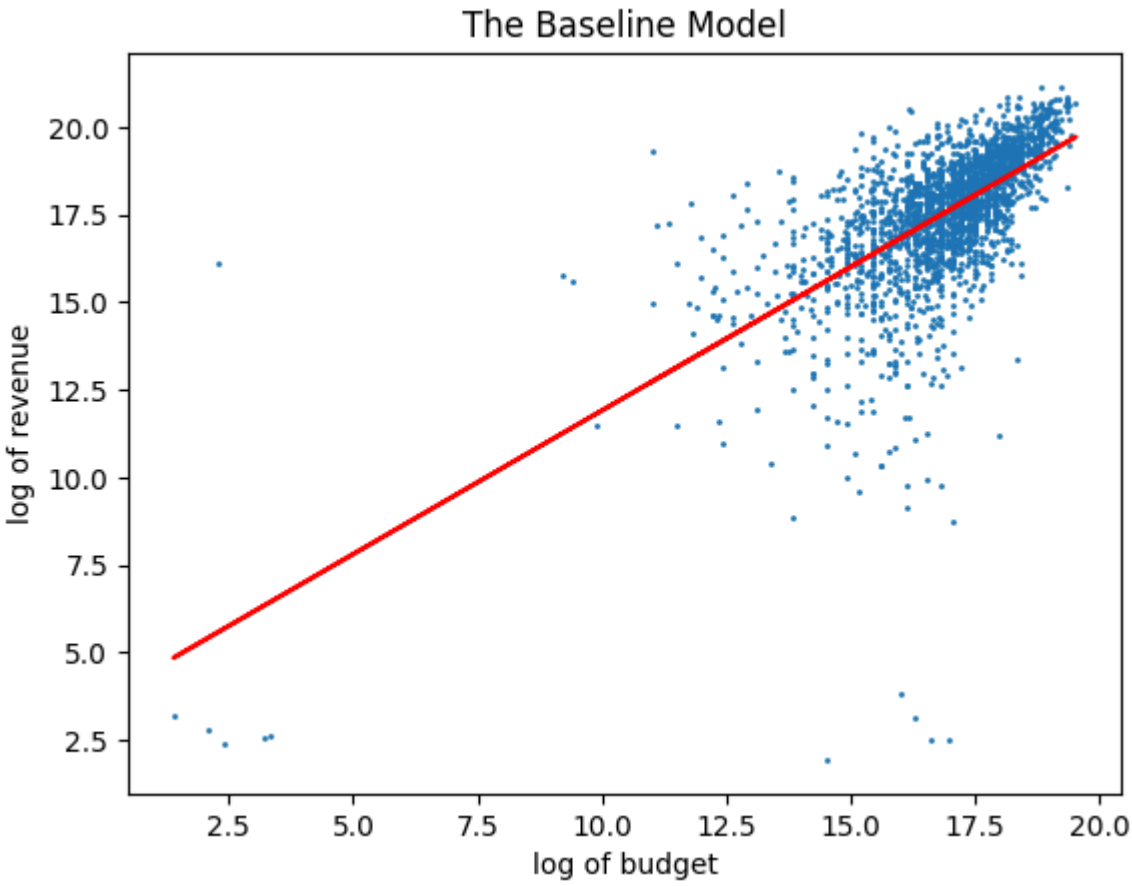
The following model will be a baseline model for predicting movie revenue by using only one feature, budgets.

Equation: y = 0.8202 * x + 3.7123

RSE using training data : 1.5715

In [144…
```
# use log to normalize data
trainingMovies['logBudget'] = np.log(trainingMovies['budget'])
trainingMovies['logRevenue'] = np.log(trainingMovies['revenue'])

x = trainingMovies['logBudget']
y = trainingMovies['logRevenue']

slope, intercept, r, p, std_err = stats.linregress(x, y)

def myfunc(x):
    return slope * x + intercept

# linear regression equation using scipy stats
print("Equation: ")
print("y = ", str(slope), "* x", str(intercept))
mymodel = list(map(myfunc, x))

# The data graph including the linear regression function
plt.scatter(x, y, s = 1)
plt.plot(x, mymodel, color = "red")
plt.xlabel("log of budget")
plt.ylabel("log of revenue")
plt.title("The Baseline Model")
plt.show()

# Calculate RSE using traningMovies
prediectedY = myfunc(x)
n = len(y)
RSE = np.sqrt(np.sum((y - prediectedY)**2) / (n - 2))
print("RSE:", RSE)
```

```
Equation:
y =  0.8201548222806377 * x 3.712439557949196
```




```
RSE: 1.5714900813703614
```

5.2.1 Test the baseline model

Since the equation of the baseline model has been computed, it is possible to evaluate how model is accurate by comparing the actual revenues with the revenues predicted by the model.

RSE using testing data: 1.4723

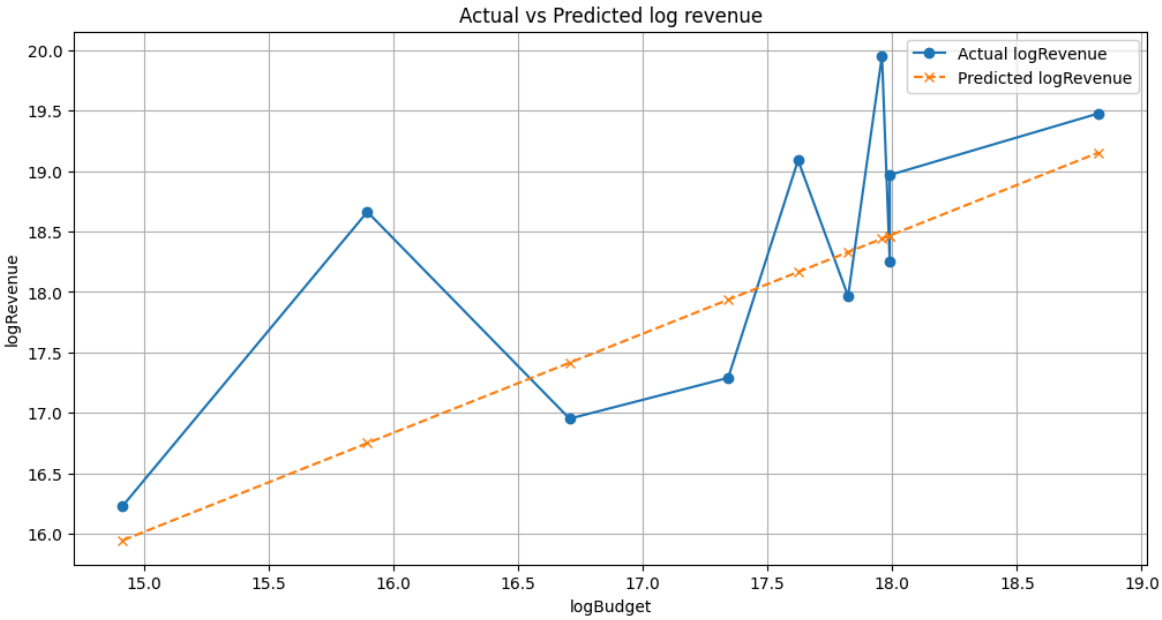
The result is similar from the result of the simple linear regression model in Case 1. Since this is a linear model, it is clear that the actual and expected revenue form a noticeable gap.

In [145…

```
# movies released after 2010.
testingMovies

# use log to normalize datas
testingMovies['logBudget'] = np.log(testingMovies['budget'])
testingMovies['logRevenue'] = np.log(testingMovies['revenue'])


y_tested = myfunc(testingMovies['logBudget'])
y_tested
testingMovies['logRevenue']

# RSE using testing data
n = len(y_tested)
RSE = np.sqrt(np.sum((y_tested - testingMovies['logRevenue'])**2) / (n - 2))
print("RSE: ", RSE)
```

```
testingMovies['predicted_logRevenue'] = y_tested
top10Movies = testingMovies.head(10)
top10Movies = top10Movies.sort_values(by = 'logBudget')

plt.figure(figsize=(12, 6))
plt.plot(top10Movies['logBudget'], top10Movies['logRevenue'], label='Actual logR
plt.plot(top10Movies['logBudget'], top10Movies['predicted_logRevenue'], label='P
plt.title('Actual vs Predicted log revenue')
plt.xlabel('logBudget')
plt.ylabel('logRevenue')
plt.legend()
plt.grid(True)
plt.show()
```

```
RSE:  1.4722642507679031
```



5.2.2 Train Multiple Linear Regression model

Based on the previous data analysis, the heatmap showed that movie revenue is correlated not only with budget but also with several other features.
Therefore, the multiple linear regression model will be used to estimate the revenue more accrurately, using multiple features.

RSE using testing data: 1.2195

In [146…

```
# take log of meanRevenuePerCountry to normalize the data
trainingMovies['logMeanRevenuePerCountry'] = np.log(trainingMovies['meanRevenueP
trainingMovies['logMeanRevenuePerCompany'] = np.log(trainingMovies['meanRevenueP

testingFeatures = ['logBudget', 'popularity', 'vote_count', 'numberOfCrews', 'lo
x = trainingMovies[testingFeatures]
y = trainingMovies['logRevenue']

regr = linear_model.LinearRegression()
regr.fit(x, y)

print("Weights (w_i):", regr.coef_)
print("Intercept (b):", regr.intercept_)
```

```
y_pred = regr.predict(x)
RSE = np.sqrt(np.sum((y - y_pred)**2) / (len(y) - 6 - 1))
print("RSE: ", RSE)
```

```
Weights (w_i): [4.35727292e-01 2.84434175e-03 2.11255595e-04 1.43318432e-03
 2.44930312e-01 7.29667980e-01]
Intercept (b): -8.062627364037464
RSE:  1.219510619982789
```

5.2.2 Test the multiple regression model

Since the multiple regression model has been trained, it is possible to evaluate how model is accurate by comparing the actual revenues with the revenues predicted by the model.

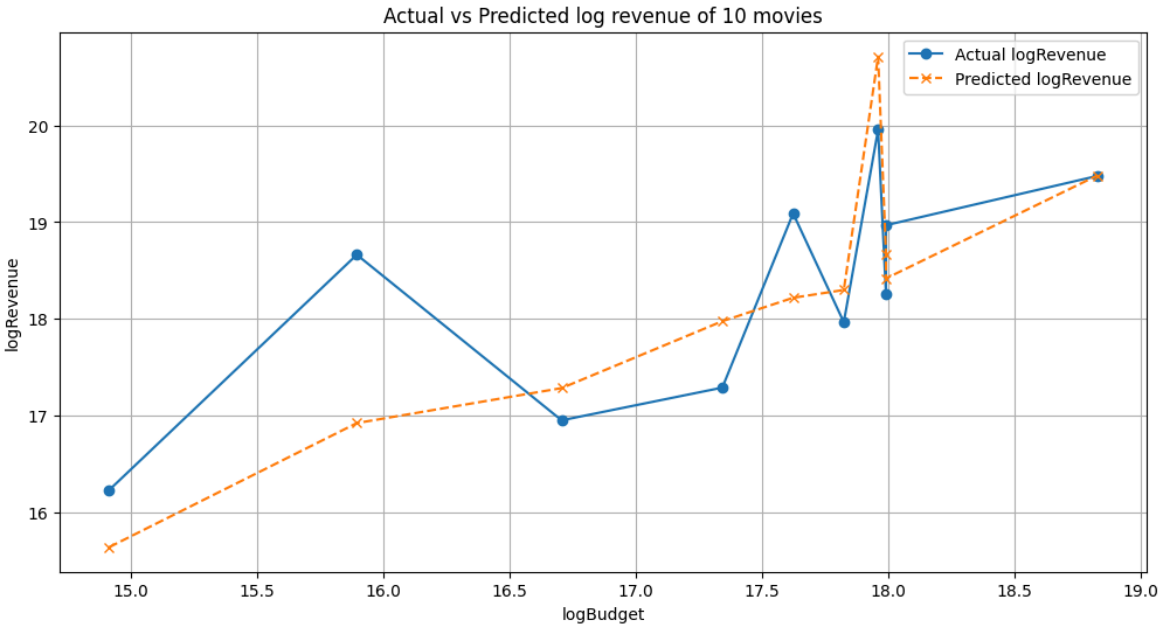
RSE using testing data: 1.1299

The RSE gap between the training and testing data becomes smaller in the random split when compared to Case 1.
Since the training and testing data are not biased toward any specific time period or a group in Case 2,
it leads to more balanced and consistent RSE results.

In [147…

```
# movies released after 2010.
testingMovies

# take log of meanRevenuePerCountry to normalize the data
testingMovies['logMeanRevenuePerCountry'] = np.log(testingMovies['meanRevenuePer
testingMovies['logMeanRevenuePerCompany'] = np.log(testingMovies['meanRevenuePer

testingY_pred = regr.predict(testingMovies[testingFeatures])

# RSE using testing data
n = len(testingY_pred)
RSE = np.sqrt(np.sum((testingY_pred - testingMovies['logRevenue'])**2) / (n - 6
print("RSE: ", RSE)

# compare acutal vs predicted log revenue
testingMovies['predicted_logRevenue'] = testingY_pred
top10Movies = testingMovies.head(10)
top10Movies = top10Movies.sort_values(by = 'logBudget')

plt.figure(figsize=(12, 6))
plt.plot(top10Movies['logBudget'], top10Movies['logRevenue'], label='Actual logR
plt.plot(top10Movies['logBudget'], top10Movies['predicted_logRevenue'], label='P
plt.title('Actual vs Predicted log revenue of 10 movies')
plt.xlabel('logBudget')
plt.ylabel('logRevenue')
plt.legend()
plt.grid(True)
plt.show()
```

```
RSE:  1.1299491393214658
```



5.2.3 Train Polynomial Regression model

Since we have already estimated movie revenues using a multiple linear regression model with several features, we now investigate whether a polynomial relationship between revenue and features produces better results.
Therefore, a polynomial regression model will be employed to estimate the revenue more accurately using the same set of features.

A degree of 2 will be used.

Alpha of 2500 will be used.

RSE using training data:

- 1.1206

In [148…

```
x = trainingMovies[['logBudget', 'popularity', 'vote_count', 'numberOfCrews', 'l
y = trainingMovies['logRevenue']

poly = PolynomialFeatures(degree = 2, include_bias = False)

# create feature matrix
scaler = StandardScaler()
scaledX = scaler.fit_transform(x)
x = poly.fit_transform(scaledX)

# train models
model = Ridge(alpha = 1)
model.fit(x, y)

# collect predicted results
y_predicted = model.predict(x)

result = pd.DataFrame({
    'logBudget' : trainingMovies['logBudget'],
    'y_predicted' : y_predicted
})
```

```
# residual sum of squares (RSS)
n = len(y)
RSE = np.sqrt(np.sum((y - y_predicted) ** 2) / (n - 6 - 1))
print("RSE:", RSE)
```

```
RSE: 1.1205604798267783
```

5.2.3 Test the polynomial regression model

Since the polynomial regression model has been trained, it is possible to evaluate how model is accurate by comparing the actual revenues with the revenues predicted by the model.

RSE using testing data: 1.1961

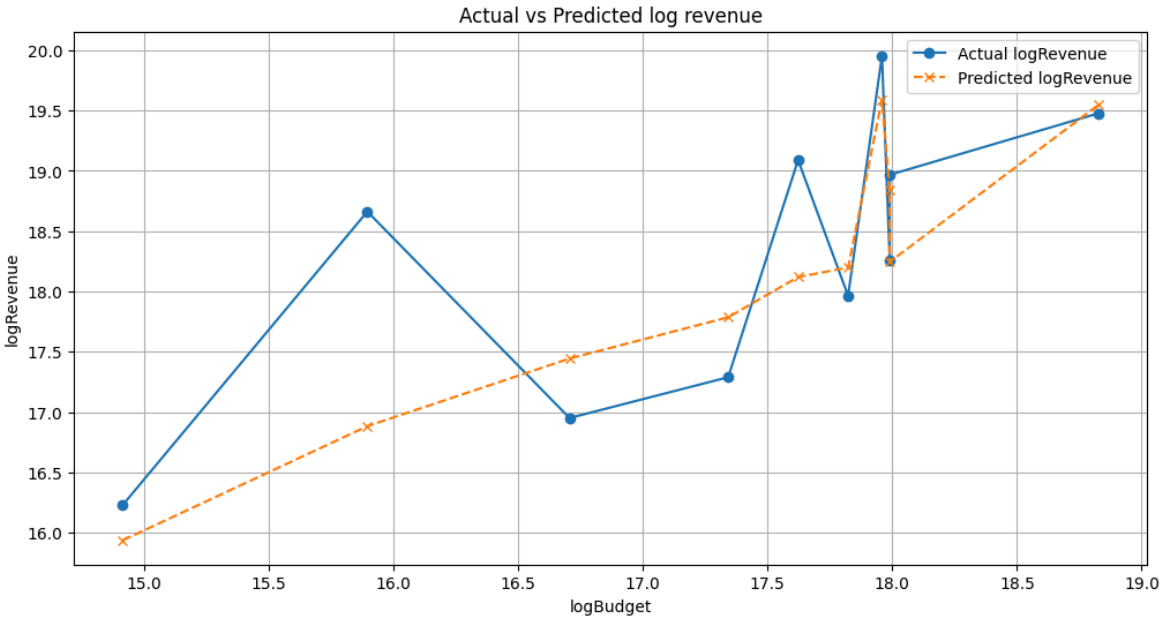
The RSE using polynomial regression is smaller than the RSE of linear regression when observing the training data, like Case 1.

Conversely, the RSE using polynomial regression is higher than the RSE of linear regression when observing the testing data, like Case 1.

In [149…

```
# movies released after 2010.
testingMovies

x = testingMovies[['logBudget', 'popularity', 'vote_count', 'numberOfCrews', 'lo
y = testingMovies['logRevenue']

scaledX = scaler.transform(x)
x = poly.transform(scaledX)

testingY_pred = model.predict(x)
# print(testingY_pred)

# RSE using testing data
n = len(testingY_pred)
RSE = np.sqrt(np.sum((y - testingY_pred)**2) / (n - 6 - 1))
print("RSE: ", RSE)


testingMovies['predicted_logRevenue'] = testingY_pred
top10Movies = testingMovies.head(10)
top10Movies = top10Movies.sort_values(by = 'logBudget')

plt.figure(figsize=(12, 6))
plt.plot(top10Movies['logBudget'], top10Movies['logRevenue'], label = 'Actual lo
plt.plot(top10Movies['logBudget'], top10Movies['predicted_logRevenue'], label =
plt.title('Actual vs Predicted log revenue')
plt.xlabel('logBudget')
plt.ylabel('logRevenue')
plt.legend()
plt.grid(True)
plt.show()
```

```
RSE:  1.195487457148468
```



Result (RSE)

The RSE decreased as we used more complex models while training.
It is reasonable that the RSE becomes smaller as the models use more complex calculations.

However, the lowest RSE of test data came from the linear regression model.
It is conjectured that this is because complex calculations were fit to the training data, so that polynomial regression resulted in a higher error for the testing data compared to the linear regression model.

In [150…

```
rseOfTrainingData = {
    "Model": ["Baseline Model", "Multiple Linear Regression", "Polynomial Regres
    "Split Before/After 2010": [1.5531, 1.1953, 1.0803],
    "Polynomial Reression": [1.5715, 1.2195, 1.1206]
}
pd.DataFrame(rseOfTrainingData)
```

Out[150…

	Model	Split Before/After 2010	Polynomial Reression
0	Baseline Model	1.5531	1.5715
1	Multiple Linear Regression	1.1953	1.2195
2	Polynomial Regression	1.0803	1.1206

In [151…

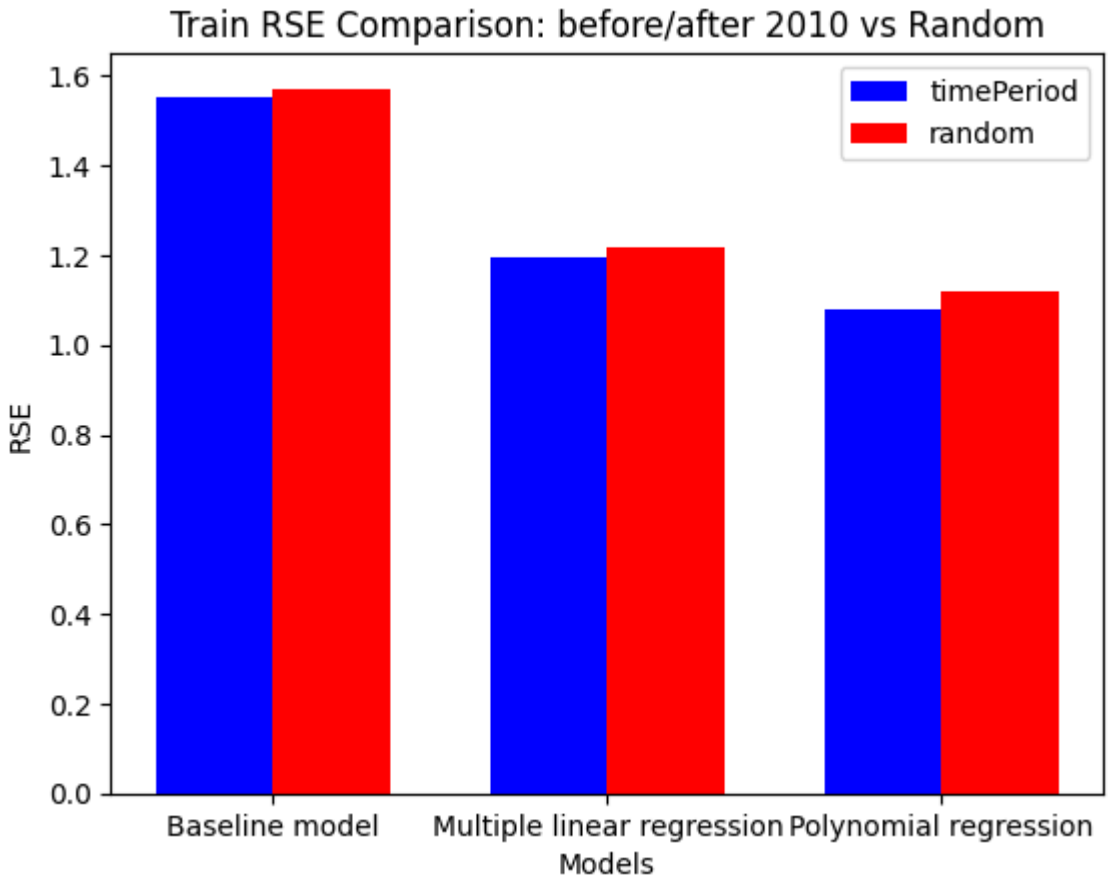
```
resOfTestingData = {
    "Model": ["Baseline Model", "Multiple Linear Regression", "Polynomial Regres
    "Split Before/After 2010": [1.5123, 1.4588, 1.9158],
    "Polynomial Reression": [1.4723, 1.1299, 1.1961]
}
pd.DataFrame(resOfTestingData)
```

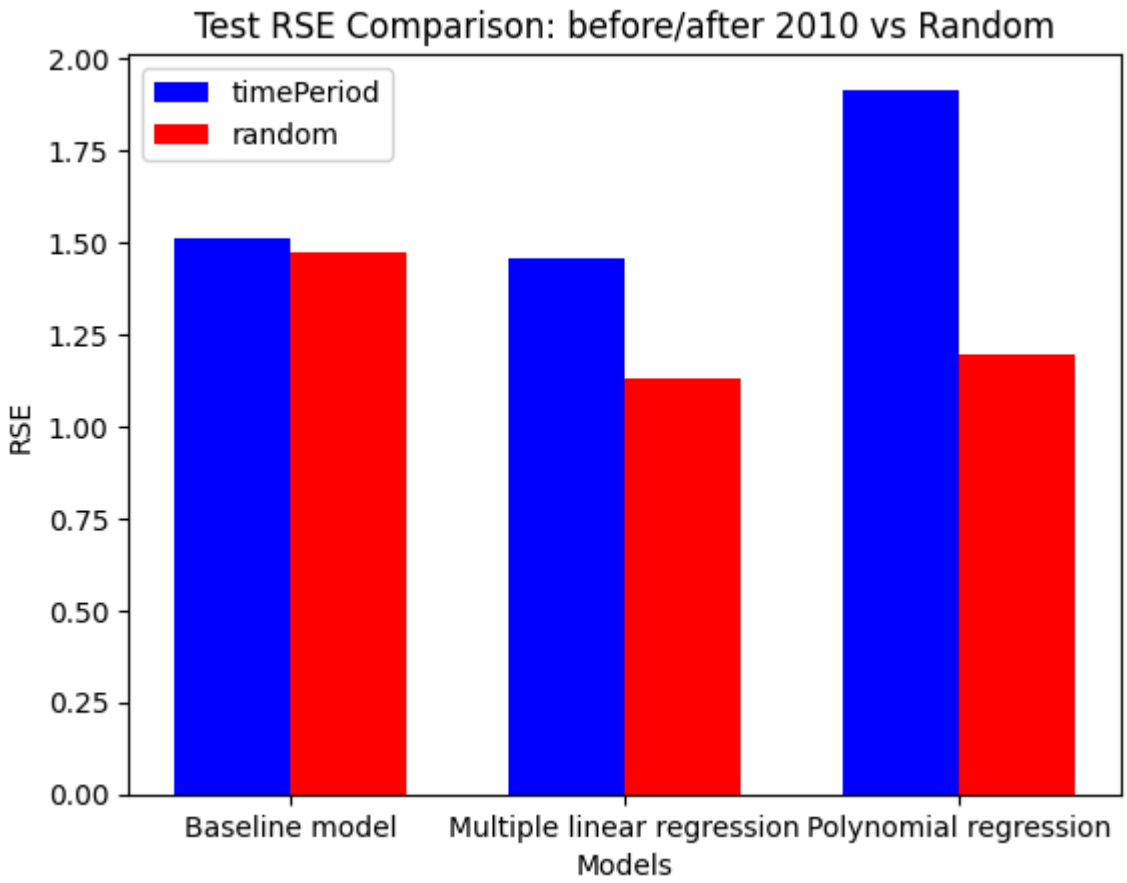
Out[151…

	Model	Split Before/After 2010	Polynomial Reression
0	Baseline Model	1.5123	1.4723
1	Multiple Linear Regression	1.4588	1.1299
2	Polynomial Regression	1.9158	1.1961

In [152…

```
label = ['Baseline model', 'Multiple linear regression', 'Polynomial regression'
trainingRseByTimePeriod = [1.5531, 1.1953, 1.0803]
trainingRseByRandom = [1.5715, 1.2195, 1.1206]

testingRseByTimePeriod = [1.5123, 1.4588, 1.9158]
testingRseByRandom = [1.4723, 1.1299, 1.1961]

# cross validation with the training RSE.
x = np.arange(len(label))
barWidth = 0.35
plt.bar(x - barWidth / 2, trainingRseByTimePeriod, width = barWidth, label = 'ti
plt.bar(x + barWidth / 2, trainingRseByRandom, width = barWidth, label = 'random
plt.xticks(x, label)
plt.xlabel('Models')
plt.ylabel('RSE')
plt.title('Train RSE Comparison: before/after 2010 vs Random')
plt.legend()
plt.show()


# cross validation with the testing RSE.
plt.bar(x - barWidth / 2, testingRseByTimePeriod, width = barWidth, label = 'tim
plt.bar(x + barWidth / 2, testingRseByRandom, width = barWidth, label = 'random'
plt.xticks(x, label)
plt.xlabel('Models')
plt.ylabel('RSE')
plt.title('Test RSE Comparison: before/after 2010 vs Random')
plt.legend()
plt.show()
```





Conlcusion

1. Data Splitting Strategy significantly influences performance

- Case 1:

When the data was divided by decades, the predicted curve followed a similar overall trend to the actual data,
the predicted values were not close to the true values.
It failed to capture the changes in the scale of the film industry and the actual growth of the market over a certain amount of time.
This indicates a limitation in analyzing the problem using only the given dataset.

- Case 2:

On the other hand, the prediction using randomly split data was not affected by time, but was made only with the data itself.
This derived a result having low RSE, though the tendency of case 1 seemed better than the one of case 2.

2. Multiple Linear regression record the lowest overall RSE

At first, we thought that the polynomial regression model would have make the best prediction
since it does the most complicated calculations among the three models. However it gave us a result worse than the multiple linear regression model because it was vulnerable to overfitting.

3. Future Improvement

In addition to movie data, time related information such as theater attendance numbers and Netflix
subscriber counts would likely lead to more accurate results. Applying a more stable feature normalization method is necessary to achieve more accurate results in polynomial regression.